**40 single-best-answer questions | 60 minutes | 40 marks**

Answer all questions. Choose one answer per question. Use a separate answer sheet.

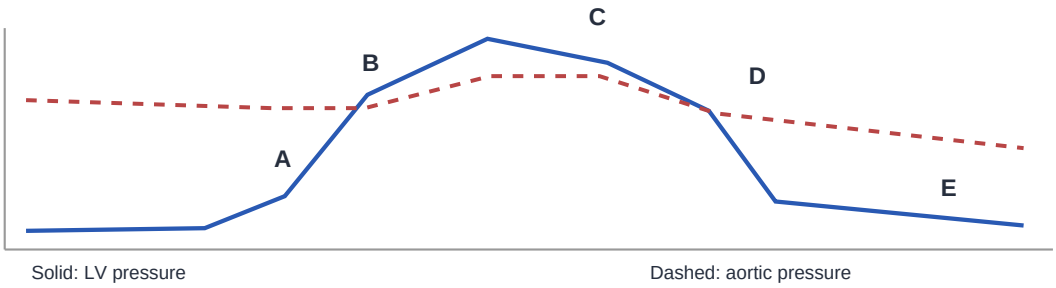
**1. Why does the sinoatrial (SA) node normally determine the heart rate?**

A. It is the only cardiac site that can generate impulses without nerve stimulation.
B. It reaches the threshold for spontaneous discharge sooner than other pacemakers.
C. It conducts electrical impulses through ventricular muscle faster than Purkinje fibres.
D. Its cells have a more negative resting membrane potential than ventricular cells.
E. It receives stronger sympathetic input than any other pacemaker tissue.

**2. Left ventricular pressure has risen to 45 mmHg. Left atrial pressure is 10 mmHg and aortic pressure is 85 mmHg. Which phase is most likely?**

A. Rapid ventricular filling
B. Reduced ventricular ejection
C. Isovolumetric relaxation
D. Rapid ventricular ejection
E. Isovolumetric contraction

**3. The diagram shows left ventricular pressure (solid) and aortic pressure (dashed). At which labelled point does aortic valve opening begin?**




A. Point B
B. Point A
C. Point C
D. Point D
E. Point E

**4. A ventricle has an end-diastolic volume of 140 mL and an end-systolic volume of 56 mL. What is its ejection fraction?**

A. 40%
B. 60%
C. 50%
D. 65%
E. 84%

**5. In complete AV dissociation, an atrium sometimes contracts while the tricuspid valve is closed. Which jugular venous finding may appear?**

A. A persistently absent a wave
B. Intermittent giant v waves caused by atrial systole
C. Loss of the y descent with every normal beat
D. A regular arterial-type pulse in the jugular vein
E. Intermittent giant a waves

**6. Why does inspiration normally widen the split between the aortic and pulmonary components of the second heart sound?**

A. Increased right ventricular filling delays mitral valve opening.
B. Increased right ventricular filling delays pulmonary valve closure.
C. Reduced right ventricular filling delays aortic valve closure.
D. Increased left ventricular filling delays tricuspid valve closure.
E. Reduced left ventricular filling delays the first heart sound.

**7. A patient has backward blood flow from the left ventricle into the left atrium during systole. Which valve disorder best explains this?**

A. Mitral stenosis
B. Aortic stenosis
C. Aortic regurgitation
D. Mitral regurgitation
E. Tricuspid regurgitation

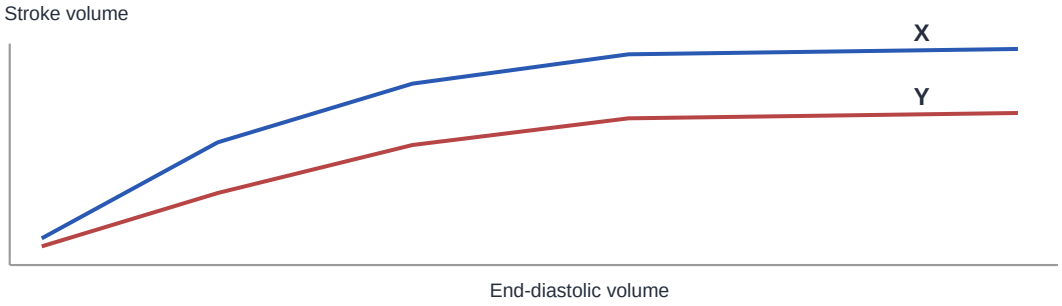
**8. Venous return rises while heart rate, afterload and contractility remain unchanged. What immediate change is expected in a normal ventricle?**

A. A rise in end-diastolic volume and stroke volume
B. A fall in end-diastolic volume and stroke volume
C. A rise in stroke volume with a fall in end-diastolic volume
D. A fall in stroke volume despite a rise in end-diastolic volume
E. No change in stroke volume despite greater ventricular filling

**9. Aortic pressure suddenly rises. For the first affected beat, ventricular filling and contractility are unchanged. What is the likely effect on stroke volume?**

A. Stroke volume increases because aortic recoil becomes stronger.
B. Stroke volume increases because ventricular filling has increased.
C. Stroke volume stays constant because heart rate has not changed.
D. Stroke volume decreases because the ventricle ejects against greater pressure.
E. Stroke volume increases because the AV node conducts more quickly.

**10. Two ventricular function curves are shown. At the same end-diastolic volume, curve X produces a larger stroke volume than curve Y. What best explains the difference?**




A. Ventricular preload must be greater on curve X.
B. The heart rate must be lower on curve X.
C. The aortic valve must be closed throughout ejection on curve X.
D. The ventricular muscle must be less responsive to calcium on curve X.
E. Ventricular contractility is greater on curve X.

**11. Oxygen consumption is 280 mL/min. Arterial oxygen content is 200 mL/L and mixed venous oxygen content is 130 mL/L. What is cardiac output?**

A. 4 L/min
B. 2 L/min
C. 3 L/min
D. 5 L/min
E. 7 L/min

**12. At the same stroke volume and heart rate, which change generally increases myocardial oxygen demand more?**

A. Pumping against a lower arterial pressure
B. Reducing the pressure needed to open the aortic valve
C. Pumping against a much higher arterial pressure
D. Lowering wall tension during ventricular ejection
E. Reducing the pressure difference across the aortic valve

**13. According to Poiseuille's law, if a vessel radius is reduced to half while other factors remain constant, how does resistance change?**

A. It becomes 2 times greater.
B. It becomes 4 times greater.
C. It becomes 16 times greater.
D. It becomes 8 times greater.
E. It becomes 16 times smaller.

**14. Why is average blood-flow velocity normally much lower in capillaries than in the aorta?**

A. The total capillary cross-sectional area is smaller than that of the aorta.
B. Capillaries receive a smaller total blood flow than the aorta in steady state.
C. Capillary pressure is lower, so red cells are stationary between heartbeats.
D. The combined cross-sectional area of all capillaries is much larger.
E. Blood viscosity is so much higher in capillaries that flow velocity must be zero.

**15. Which isolated change increases the tendency for turbulent flow in a blood vessel?**

A. An increase in blood viscosity
B. A decrease in blood density
C. An increase in blood-flow velocity
D. A decrease in vessel diameter
E. A decrease in the Reynolds number

**16. A resting adult has a blood pressure of 120/75 mmHg. Using the usual resting approximation, what is mean arterial pressure?**

A. 75 mmHg
B. 90 mmHg
C. 85 mmHg
D. 95 mmHg
E. 105 mmHg

**17. In a healthy adult, cuff pressure is slowly released during blood-pressure measurement. Disappearance of Korotkoff sounds usually marks which pressure?**

A. Systolic arterial pressure
B. Mean arterial pressure
C. Diastolic arterial pressure
D. Pulse pressure
E. Central venous pressure

**18. A deep vein in one leg becomes obstructed, increasing pressure in vessels upstream. What local change most directly favours swelling?**

A. An increase in plasma albumin concentration
B. A decrease in capillary hydrostatic pressure
C. An increase in lymphatic drainage capacity
D. A decrease in capillary wall permeability
E. An increase in capillary hydrostatic pressure

**19. A patient develops chronic swelling of an arm after axillary lymph nodes are removed. What is the main reason?**

A. Arterial blood supply to the arm has completely stopped.
B. Red blood cells begin moving freely out of intact capillaries.
C. The heart pumps excess blood only into the affected arm.
D. Interstitial fluid and proteins cannot drain normally through lymph vessels.
E. Plasma albumin production increases sharply in the affected tissue.

**20. Why does walking help return blood from the leg veins to the heart?**

A. Muscle contractions compress veins while valves limit backward flow.
B. Venous valves in the calf actively push blood toward the heart.
C. Walking lowers all leg capillary pressures to zero.
D. Muscle compression pushes venous blood toward the feet through open valves.
E. Veins become less compliant and no longer experience gravity.

**21. During exercise, working skeletal muscle produces more $CO_2$, $H^+$ and other metabolites. What local vascular response is expected?**

A. Arteriolar constriction that reduces supply to active muscle
B. Arteriolar dilation that increases blood supply to the active muscle
C. Closure of all capillaries until the exercise finishes
D. A reduction in local flow despite increased oxygen demand
E. Selective closure of venous valves inside the muscle

**22. Blood flow through a forearm artery is briefly blocked and then restored. Why may flow temporarily rise above its resting level?**

A. Vasodilator metabolites built up while the tissue was underperfused.
B. The heart has permanently increased the forearm's blood supply.
C. The arterial pulse wave has stopped travelling through the arm.
D. The blockage permanently removed vascular resistance.
E. Venous valves now actively force arterial blood toward the hand.

**23. The graph shows blood flow across a range of perfusion pressures. Between points B and C, why is flow nearly constant despite rising pressure?**

A. Arteriolar resistance falls as perfusion pressure rises.
B. Cardiac output stops changing throughout the entire body.
C. Arteriolar resistance rises as perfusion pressure rises.
D. Venous blood begins flowing backward into the artery.
E. The tissue stops consuming oxygen over this pressure range.

**24. A person suddenly loses blood and arterial pressure falls. Which immediate autonomic change helps restore pressure?**

A. Reduced sympathetic activity and increased vagal activity
B. Reduced sympathetic and reduced vagal activity together
C. Increased sympathetic activity and reduced vagal activity
D. Increased vagal activity with arteriolar dilation
E. No change in autonomic outflow until blood volume recovers

**25. A patient develops rapidly rising intracranial pressure with impaired brainstem perfusion. Which combination is typical of the Cushing response?**

A. Increased arterial pressure with reflex slowing of the heart
B. Decreased arterial pressure with reflex slowing of the heart
C. Increased arterial pressure with sustained reflex tachycardia
D. Decreased arterial pressure with complete loss of sympathetic tone
E. Unchanged arterial pressure with marked peripheral vasodilation

**26. Rapid infusion of fluid increases venous return and stretches the right atrium. Which reflex response may help accommodate the larger venous return?**

A. Heart rate may decrease because atrial stretch directly increases vagal outflow.
B. Heart rate may increase as atrial stretch activates a reflex.
C. Heart rate may remain unchanged because atrial stretch has no reflex effects.
D. Heart rate may decrease because venous return is reduced by the fluid infusion.
E. Heart rate may increase only if the arterial baroreceptors first detect hypotension.

**27. Immediately after a person stands up quickly, before reflex compensation has fully developed, what tends to happen?**

A. Venous pooling decreases and ventricular filling rises.
B. Venous pooling increases and venous return to the heart falls.
C. Venous pressure in the feet falls below pressure at heart level.
D. Cardiac output rises because gravity pushes blood toward the chest.
E. Cerebral blood flow increases because the head is now above the heart.

**28. A person loses a substantial amount of blood. What sequence most directly explains the initial fall in cardiac output?**

A. Lower blood volume directly raises ventricular end-diastolic volume.
B. Lower blood volume immediately increases plasma oncotic pressure enough to raise output.
C. Lower blood volume causes the aortic valve to open earlier at a higher stroke volume.
D. Lower blood volume leads to lower venous return and ventricular filling.
E. Lower blood volume raises pulmonary venous return above systemic venous return.

**29. A patient with severe infection has widespread arteriolar dilation and low arterial pressure. Which shock mechanism is most likely?**

A. Hypovolaemic shock caused mainly by external blood loss
B. Cardiogenic shock caused mainly by poor ventricular pumping
C. Obstructive shock caused mainly by mechanical blockage of blood flow
D. A normal circulation with a temporary rise in vascular resistance
E. Distributive shock caused mainly by reduced systemic vascular resistance

**30. A patient with left ventricular failure develops breathlessness and pulmonary oedema. What pressure change most directly promotes fluid movement into the lungs?**

A. Increased pulmonary venous and capillary hydrostatic pressure
B. Decreased pulmonary venous pressure
C. Increased plasma oncotic pressure in pulmonary capillaries
D. Decreased hydrostatic pressure in pulmonary capillaries
E. Increased lymphatic drainage with unchanged capillary pressure

**31. An obstruction mainly affects blood supply to the lateral surface of a cerebral hemisphere. Which major cerebral artery is most likely involved?**

A. Anterior cerebral artery
B. Posterior cerebral artery
C. Anterior communicating artery
D. Middle cerebral artery
E. Basilar artery

**32. The cerebral aqueduct becomes blocked while CSF production continues. Which ventricles are expected to enlarge first?**

A. The fourth ventricle alone
B. The third and fourth ventricles only
C. The lateral ventricles and fourth ventricle only
D. The lateral and third ventricles
E. The fourth ventricle and central canal only

**33. Mean arterial pressure is 95 mmHg and intracranial pressure is 15 mmHg. Assuming cerebral venous pressure is not higher than intracranial pressure, what is cerebral perfusion pressure?**

A. 65 mmHg
B. 75 mmHg
C. 80 mmHg
D. 95 mmHg
E. 110 mmHg

**34. Why does most left ventricular coronary blood flow occur during diastole rather than systole?**

A. The left coronary artery is closed by the aortic valve during systole.
B. Blood in coronary arteries contains oxygen only during diastole.
C. The heart stops using oxygen during systole.
D. Pulmonary venous pressure becomes higher than aortic pressure during systole.
E. Contracting left ventricular muscle compresses vessels within its wall.

**35. One group of alveoli is poorly ventilated while nearby alveoli are well ventilated. What happens to the small pulmonary arteries supplying the poorly ventilated region?**

A. They dilate, directing more blood toward the poorly ventilated alveoli.
B. They remain unchanged because oxygen never affects pulmonary resistance.
C. They constrict only if systemic arterial pressure is also high.
D. They close because all pulmonary arteries normally lack smooth muscle.
E. They constrict, directing more blood toward better-ventilated alveoli.

**36. During exercise, pulmonary blood flow increases greatly but pulmonary arterial pressure rises relatively little. What helps explain this?**

A. Most pulmonary capillaries close as flow increases.
B. Previously underperfused capillaries open and existing vessels widen.
C. Pulmonary vascular resistance increases in direct proportion to flow.
D. The right ventricle stops pumping blood during exercise.
E. The alveoli stop exchanging gases while blood flow rises.

**37. After firm stroking of the skin, redness spreads beyond the directly stimulated line. What produces this surrounding flare?**

A. An axon reflex causing nearby arterioles to dilate
B. Clotting of capillaries at the site of injury
C. A sudden fall in total circulating blood volume
D. Closure of all precapillary sphincters around the area
E. Direct filtration of red blood cells into the epidermis

**38. What change occurs in the pulmonary circulation with the newborn's first breaths?**

A. Pulmonary vascular resistance rises as alveolar oxygen rises.
B. Pulmonary blood flow stops because placental blood flow has ended.
C. Pulmonary vascular resistance falls as the lungs expand and oxygen rises.
D. The ductus venosus begins carrying blood from the right ventricle.
E. Systemic vascular resistance falls because the placenta remains connected.

**39. Which pair of electrodes forms standard bipolar lead I in an ECG?**

A. Right arm (negative) and left leg (positive)
B. Left arm (negative) and left leg (positive)
C. Left leg (negative) and right arm (positive)
D. Right arm (negative) and left arm (positive)
E. Left arm (negative) and right arm (positive)

**40. In the ECG strip, the distance between consecutive R waves is 20 small squares. The paper speed is 25 mm/s. What is the heart rate?**

A. 60 beats/min
B. 75 beats/min
C. 80 beats/min
D. 100 beats/min
E. 120 beats/min